University of Management and Technology

School of Systems and Technology

Department of Informatics and Systems

Database Systems **Class Activity # 1** **Instructor: Ms. Anam Umera**

Total marks: 30 **CLO-1**

Note: Please submit a soft copy of the activity on LMS. Submission will not be accepted via email or any other means.

Q1: Data vs Information (Scenario)

Scenario:
A teacher collects the following marks of students in a quiz:
65, 70, 80, 90, 75

Questions:

1. Identify whether the quiz marks given above represent **data or information**.
2. If the teacher calculates the **average marks**, what type of result will it represent?
3. What does the term **information** mean in this scenario?
4. Briefly explain the **difference between data and information** using this example.

Q2: Understanding Database (Scenario)

Scenario:
A university wants to store information about students such as **Student ID, Name, Department, Email, and GPA** in a system.

Questions:

1. Identify the type of system that should be used to store and manage this information.
2. What is meant by a **database** in this scenario?
3. Define the term **Database Management System (DBMS)**.
4. List two examples of **student data** that will be stored in the database.

Q3: Metadata (Scenario)

Scenario:
A student table in a university database contains the following columns:
StudentID, Name, Department, GPA

Questions:

1. Identify what the **column names** in the table represent.
2. Define the term **metadata**.
3. Give possible **data types** for StudentID and Name.
4. Explain the **importance of metadata** in a database system.

Q4: File System Problems (Scenario)

Scenario:
A small college stores student records in **separate Excel files** in different departments. Sometimes the same student's information appears in multiple files.

Questions:

1. Identify the problem that occurs when the same data is stored in multiple files.
2. What is meant by **data inconsistency** in this situation?
3. Briefly explain why maintaining data in separate files can cause difficulties.
4. State how a **Database Management System (DBMS)** can help solve this problem.

Q5: Database Users/Roles (Scenario)

Scenario:
In a university database system:

- The **IT administrator** manages the database.
- **Teachers** enter student marks.
- **Students** view their results.

Questions:

1. Identify the **Database Administrator (DBA)** in this scenario.
2. Who are considered the **end users** of the database system?
3. What role do **teachers** perform in the database system?
4. Explain why different users have **different roles** in a database environment.